A CRITERION FOR RELATIVE AMPLENESS IN TERMS OF IMMERSIONS INTO PROJECTIVE BUNDLES, FOR FINITE TYPE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Six-line proof body carrying both directions of an equivalence; the reduction steps (affine covering of the base, application of the affine criterion, application of the relative criterion) are stated, so the span is a proof and not a pointer.

Allowed standard prerequisites (editorial).

- The affine criterion for ampleness and very ampleness of finite type morphisms. Lemma \ref{lemma-finite-type-over-affine-ample-very-ample}, cited at proof lines 9634-9636. Gives the forward implication on each affine piece.
- The affine open criterion for relative ampleness. Lemma \ref{lemma-characterize-relatively-ample}, cited at proof lines 9637-9639. Gives the reverse implication.
- The hypothesis that the morphism is of finite type. the statement's assumption, used at proof lines 9633-9636. Makes the affine criterion applicable.

Source reference. Original lemma 02NR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3490–3490. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01VT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2458–2458. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01VJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2448–2448. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{L}$ be an invertible sheaf on X. Assume f is of finite type. The following are equivalent:*

(1) *$\mathcal{L}$ is f-relatively ample, and*
(2) *there exist an open covering $S = \bigcup V_j$, for each j an integers $d_j \geq 1$, $n_j \geq 0$, and immersions*
$$i_j : X_j = f^{-1}(V_j) = V_j \times_S X \longrightarrow \mathbf{P}^{n_j}_{V_j}$$
over V_j such that $\mathcal{L}^{\otimes d_j}|_{X_j} \cong i_j^* \mathcal{O}_{\mathbf{P}^{n_j}_{V_j}}(1)$.

Proof. We see that (1) implies (2) by taking an affine open covering of S and applying Lemma 01VT to each of the restrictions of f and $\mathcal{L}$. We see that (2) implies (1) by Lemma 01VJ. □